

Appendix

Results on Temperature Optimum Dataset Across Enzyme Classes

Table 1–7 presents the detailed performance of all models across different enzyme classes (EC1 to EC7). PatchET demonstrates consistent superiority in nearly all classes, particularly in terms of MAE and RMSE.

For EC1 to EC4, PatchET achieves the lowest error values among all methods, with notable improvements over state-of-the-art baselines. For instance, in EC1, PatchET reduces MAE to 9.66 and RMSE to 13.82—substantially outperforming Transformer and Seq2Topt. In EC2, PatchET attains the highest Pearson correlation (0.545) and Spearman correlation (0.352), alongside the lowest MAE (8.17) and RMSE (11.43), indicating a particularly strong predictive capability for this class.

PatchET continues to lead in EC3 and EC4. In EC3, it achieves the highest Pearson correlation (0.562), while also reducing MAE and RMSE to 10.59 and 14.52, respectively. In EC4, PatchET shows a dramatic error reduction, with RMSE and MAE dropping to 10.94 and 8.22—clearly outperforming all other models.

While DeepET performs strongly in EC5 and EC7, PatchET still remains competitive and achieves leading results in EC6 and EC7. In EC6, PatchET records the highest Pearson correlation (0.334), lowest MAE (12.61), and lowest RMSE (17.73). Notably, in EC7, PatchET outperforms all other methods by a large margin, achieving a Pearson correlation of 0.728 and a Spearman correlation of 0.734, along with a remarkably low MAE of 5.13 and RMSE of 6.27—the best performance recorded in any enzyme class.

Overall, PatchET demonstrates robust and superior performance across the board, highlighting its generalizability and effectiveness in modeling enzyme temperature optima across diverse reaction types.

The scatter plots in Figure 1–9 offer a visual comparison of the prediction accuracy for all evaluated models. BiLSTM struggles to generalize to enzymes with optimal temperatures above 80 °C, while CNN performs better but still lacks precision across the full range. The Transformer model restricts its predictions primarily between 20 and 80 °C, failing to capture outlier behavior.

RNN yields a broader prediction distribution but tends to overestimate low-temperature enzymes. DeepET performs well for low-temperature samples but exhibits degradation at higher values. Light Attention, Seq2Topt, and TemStaPro yield more stable predictions but suffer from large errors for enzymes with low optimal temperatures.

In contrast, PatchET produces predictions that align closely with ground truth across both low and high temperature regions. It minimizes the prediction error in challenging low-temperature zones while maintaining strong performance on high-temperature enzymes.

Table 1: Results for EC1 on the Temperature Optimum Test Dataset.

Model	Pearson	Spearman	MAE	RMSE
BiLSTM	0.485	0.333	11.04	16.06
CNN	0.521	0.374	11.84	15.48
Transformer	0.547	0.367	12.43	16.25
Light Attention	0.523	0.353	11.91	15.56
RNN	0.456	0.299	13.37	18.17
DeepET	0.436	0.276	12.28	15.47
Seq2Topt	0.554	0.396	11.61	15.58
TemStaPro	0.503	0.361	12.19	15.59
PatchET	0.512	0.313	9.66	13.82

Table 2: Results for EC2 on the Temperature Optimum Test Dataset.

Model	Pearson	Spearman	MAE	RMSE
BiLSTM	0.413	0.184	10.25	15.36
CNN	0.469	0.268	10.71	14.38
Transformer	0.484	0.266	11.16	15.91
Light Attention	0.479	0.266	10.38	14.36
RNN	0.426	0.216	12.09	16.67
DeepET	0.358	0.260	11.81	14.52
Seq2Topt	0.486	0.261	10.36	14.31
TemStaPro	0.460	0.263	10.83	14.57
PatchET	0.545	0.352	8.17	11.43

Table 3: Results for EC3 on the Temperature Optimum Test Dataset.

Model	Pearson	Spearman	MAE	RMSE
BiLSTM	0.401	0.365	13.95	19.05
CNN	0.527	0.478	11.63	15.76
Transformer	0.527	0.482	12.46	16.79
Light Attention	0.530	0.473	11.68	15.59
RNN	0.469	0.419	14.48	19.21
DeepET	0.487	0.451	11.57	15.13
Seq2Topt	0.522	0.469	11.87	16.28
TemStaPro	0.519	0.476	11.66	15.71
PatchET	0.562	0.483	10.59	14.52

Table 4: Results for EC4 on the Temperature Optimum Test Dataset.

Model	Pearson	Spearman	MAE	RMSE
BiLSTM	0.443	0.214	11.13	15.55
CNN	0.353	0.185	11.83	14.84
Transformer	0.362	0.159	12.42	15.97
Light Attention	0.387	0.189	11.81	14.75
RNN	0.380	0.227	12.47	16.11
DeepET	0.362	0.265	11.07	13.59
Seq2Topt	0.408	0.214	11.85	14.94
TemStaPro	0.347	0.186	11.73	14.67
PatchET	0.526	0.440	8.22	10.94

Table 5: Results for EC5 on the Temperature Optimum Test Dataset.

Model	Pearson	Spearman	MAE	RMSE
BiLSTM	0.593	0.618	12.63	17.32
CNN	0.591	0.654	12.56	15.49
Transformer	0.618	0.671	13.42	17.21
Light Attention	0.614	0.683	11.57	14.91
RNN	0.608	0.666	12.62	16.14
DeepET	0.759	0.792	10.44	12.27
Seq2Topt	0.612	0.669	12.44	15.44
TemStaPro	0.581	0.649	12.52	15.49
PatchET	0.522	0.500	12.35	15.56

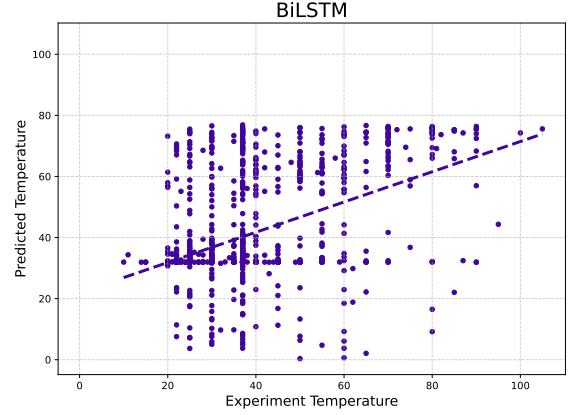


Figure 1: Scatter plot of predicted vs. actual values for the BiLSTM model.

Table 6: Results for EC6 on the Temperature Optimum Test Dataset.

Model	Pearson	Spearman	MAE	RMSE
BiLSTM	0.182	0.320	20.10	25.76
CNN	0.238	0.273	17.67	22.94
Transformer	0.250	0.207	20.38	25.69
Light Attention	0.274	0.284	18.47	23.51
RNN	0.235	0.250	21.18	26.93
DeepET	0.071	0.165	15.99	20.63
Seq2Topt	0.235	0.260	16.85	22.17
TemStaPro	0.271	0.298	17.66	22.75
PatchET	0.334	0.311	12.61	17.73

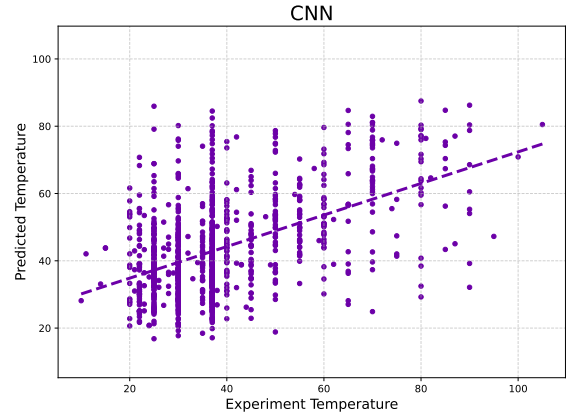


Figure 2: Scatter plot of predicted vs. actual values for the CNN model.

Table 7: Results for EC7 on the Temperature Optimum Test Dataset.

Model	Pearson	Spearman	MAE	RMSE
BiLSTM	0.570	0.094	7.83	9.48
CNN	0.265	-0.099	10.06	11.62
Transformer	0.439	-0.114	9.43	12.35
Light Attention	0.446	-0.208	9.17	11.15
RNN	0.406	0.198	8.50	12.68
DeepET	0.676	0.634	8.15	10.33
Seq2Topt	0.299	-0.074	10.25	12.05
TemStaPro	0.262	-0.173	9.83	11.57
PatchET	0.728	0.734	5.13	6.27

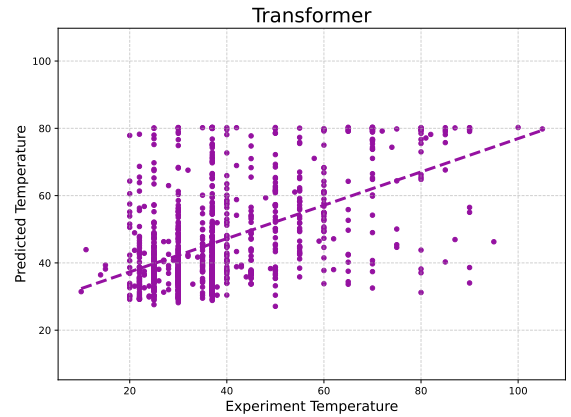


Figure 3: Scatter plot of predicted vs. actual values for the Transformer model.

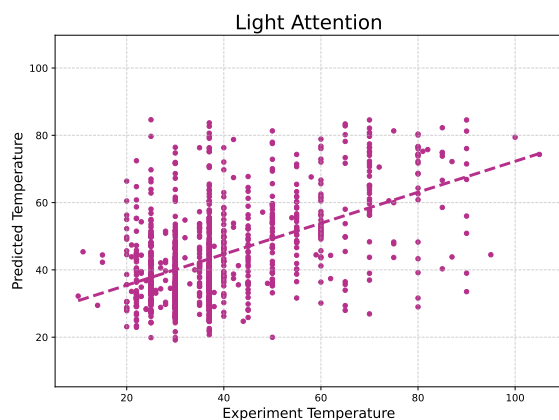


Figure 4: Scatter plot of predicted vs. actual values for the Light Attention model.

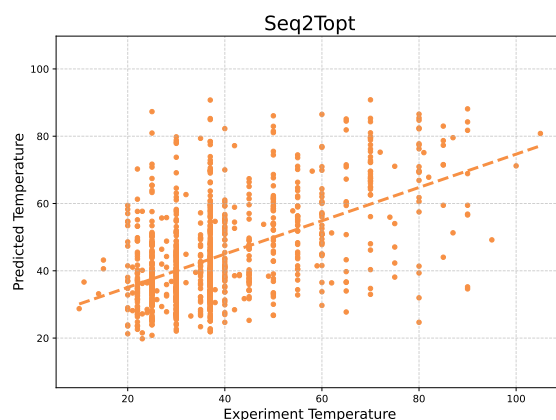


Figure 7: Scatter plot of predicted vs. actual values for the Seq2Topt model.

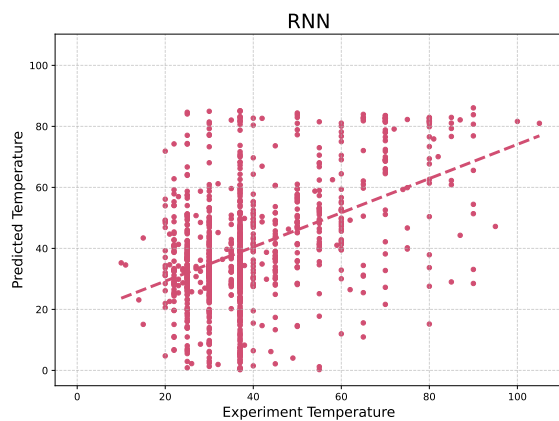


Figure 5: Scatter plot of predicted vs. actual values for the RNN model.

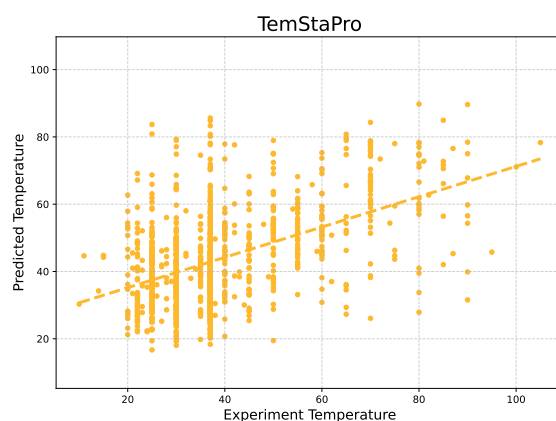


Figure 8: Scatter plot of predicted vs. actual values for the TemStaPro model.

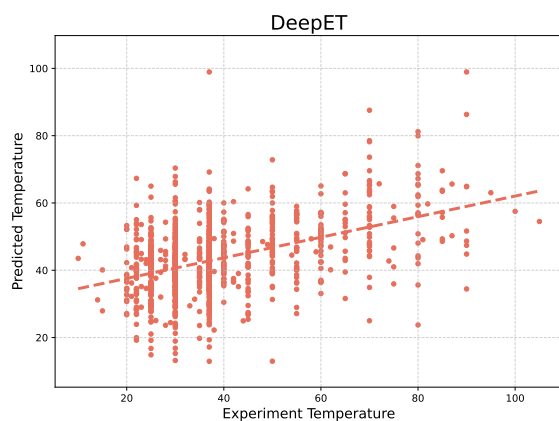


Figure 6: Scatter plot of predicted vs. actual values for the DeepET model.

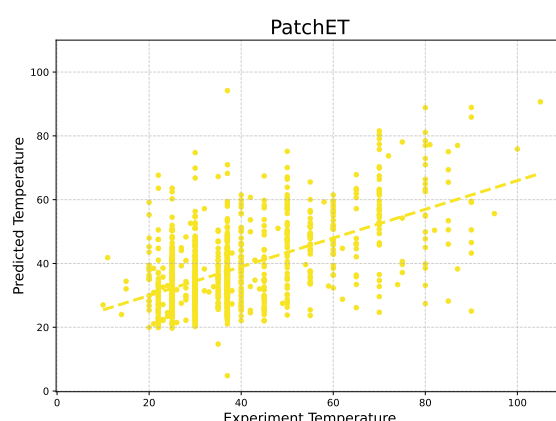


Figure 9: Scatter plot of predicted vs. actual values for the PatchET model.